

BBBBBBBBBB	UU	UU	CCCCCCCC	HH	HH	JJJJJJJJ	0000000000	BBBBBBBBBB				
BBBBBBBBBB	UU	UU	CCCCCCCC	HH	HH	JJJJJJJJ	0000000000	BBBBBBBBBB				
BB	BB	UU	UU	CC	CC	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UU	UU	CC	HH	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UU	UU	CC	HH	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UUUUUUUUUU	CCCCCCCC	HH	HH	HH	JJJJJJ	0000000000	BBBBBBBBBB				
BBBBBBBBBB	UUUUUUUUUU	CCCCCCCC	HH	HH	HH	JJJJJJ	0000000000	BBBBBBBBBB				

JJJJJJJJJ	77777777777	55555555555	222222222	AAAAAAAAA			
JJJJJJJJJ	77777777777	55555555555	22222222222	AAAAAAAAAAAA			
JJ	77	77	55	22	22	AA	AA
JJ		77	55		22	AA	AA
JJ		77	55		22	AA	AA
JJ		77	55555555		22	AAAAAAAAAAAA	
JJ		77	555555555		22	AAAAAAAAAAAA	
JJ		77		55	22	AA	AA
JJ	JJ	77		55	22	AA	AA
JJ	JJ	77		55	22	AA	AA
JJJJJJJ	77	55555555555	22222222222			AA	AA
JJJJJ	77	55555555555	22222222222			AA	AA

```
*****A  START  JOB  752  BUCHJOB  HASH TABLE  ROOM  10.28.32 PM 24 MAY 26  PRINTER1  SYS HMVS  JOB  752  START  A*****
```

J E S 2 J O B L O G

22.28.32 JOB 752 \$HASP373 BUCHJOB STARTED - INIT 1 - CLASS A - SYS HMVS
22.28.32 JOB 752 IEF403I BUCHJOB - STARTED - TIME=22.28.32
22.28.32 JOB 752 IEFACRT COB /IKFCBL00/00:00:00.08/00:00:00.14/00000/BUCHJOB
22.28.32 JOB 752 IEFACRT GO /LOADER /00:00:00.02/00:00:00.05/00000/BUCHJOB
22.28.32 JOB 752 IEF404I BUCHJOB - ENDED - TIME=22.28.32
22.28.32 JOB 752 \$HASP395 BUCHJOB ENDED

----- JES2 JOB STATISTICS -----

24 MAY 26 JOB EXECUTION DATE

126 CARDS READ

262 SYSOUT PRINT RECORDS

0 SYSOUT PUNCH RECORDS

0.00 MINUTES EXECUTION TIME

1	//BUCHJOB	JOB 1, 'HASH TABLE', CLASS=A, MSGCLASS=A,	JOB 752
2	// MSGLEVEL=(1,1), USER=SIMON, PASSWORD=		
3	***		
4	2	//COMP EXEC COBUCG2	
5	3	XXCOBUCG2 PROC CPARM1='LOAD,APOST,NODECK',	00000100
6	XX	CPARM2='SIZE=2048K,BUF=1024K'	00000200
7	***	PROC FOR COBOL 1.2.4	00000300
8	4	XXCOB EXEC PGM=IKFCBL00, REGION=1M,	00000400
9	XX	PARM='&CPARM1,&CPARM2'	00000500
10	5	XXSTEPLIB DD DSN=SYS2.VSCOLIB, DISP=SHR	00000600
11	6	XXSYSPRINT DD SYSOUT=*	00000700
12	7	XXSYSUT1 DD UNIT=SYSDA, SPACE=(460,(700,100))	00000800
13	8	XXSYSUT2 DD UNIT=SYSDA, SPACE=(460,(700,100))	00000900
14	9	XXSYSUT3 DD UNIT=SYSDA, SPACE=(460,(700,100))	00001000
15	10	XXSYSUT4 DD UNIT=SYSDA, SPACE=(460,(700,100))	00001100
16	11	XXSYSLIN DD DSN=&LOADSET, UNIT=SYSDA, DISP=(MOD,PASS),	00001200
17	XX	SPACE=(80,(500,100)), DCB=BLKSIZE=800	00001300
18	12	//SYSIN DD *	
19	13	XXGO EXEC PGM=LOADER, PARM='MAP,LET', COND=(5,LT,COB),	00001400
20	XX	REGION=2M	00001500
21	14	XXSYSLIN DD DSN=*.COB.SYSLIN, DISP=(OLD,DELETE)	00001600
22	15	XXSYSLOUT DD SYSOUT=*	00001700
23	16	XXSYSLIB DD DSN=SYS2.VSCLLIB, DISP=SHR	00001800
24	17	//GO.SYSOUT DD SYSOUT=*	
25	//		
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STMT NO. MESSAGE

```
2 4 IEF653I SUBSTITUTION JCL - PARM='LOAD,APOST,NODECK,SIZE=2048K,BUF=1024K'
4 IEF236I ALLOC. FOR BUCHJOB COB COMP
6 IEF237I 180 ALLOCATED TO STEPLIB
8 IEF237I 191 ALLOCATED TO SYSUT1
10 IEF237I 191 ALLOCATED TO SYSUT2
12 IEF237I 191 ALLOCATED TO SYSUT3
14 IEF237I 191 ALLOCATED TO SYSUT4
16 IEF237I 191 ALLOCATED TO SYSLIN
18 IEF237I JES2 ALLOCATED TO SYSIN
20 IEF142I BUCHJOB COB COMP - STEP WAS EXECUTED - COND CODE 0000
22 IEF285I SYS2.VSCOLIB KEPT *------0
24 IEF285I VOL SER NOS= PUB000.
26 IEF285I JES2.JOB00752.S00102 SYSOUT
28 IEF285I SYS26144.T222832.RA000.BUCHJOB.R0000001 DELETED *------6
30 IEF285I VOL SER NOS= MYSTUF.
32 IEF285I SYS26144.T222832.RA000.BUCHJOB.R0000002 DELETED *------6
34 IEF285I VOL SER NOS= MYSTUF.
36 IEF285I SYS26144.T222832.RA000.BUCHJOB.R0000003 DELETED *------12
38 IEF285I VOL SER NOS= MYSTUF.
40 IEF285I SYS26144.T222832.RA000.BUCHJOB.R0000004 DELETED *------6
42 IEF285I VOL SER NOS= MYSTUF.
44 IEF285I SYS26144.T222832.RA000.BUCHJOB.LOADSET PASSED *------9
46 IEF285I VOL SER NOS= MYSTUF.
48 IEF285I JES2.JOB00752.SI0101 SYSIN
50 IEF373I STEP /COB / START 26144.2228
52 IEF374I STEP /COB / STOP 26144.2228 CPU 0MIN 00.07SEC SRB 0MIN 00.01SEC VIRT 1080K SYS 224K
54 **** JOB NAME: BUCHJOB JOBCARD READ 2026/144 22:28:32 370/ 33 VS2 R03.8 HMVS *****
56 *
58 * STEP NUMBER: 1 USER CORE: 1080K START TIME: 22:28:32 CPU TIME: 00:00:00.08 ACTIVE TIME: 00:00:00.10 *
60 * STEP NAME: COB SYSTEM CORE: 224K STOP TIME: 22:28:32 SRB TIME: 00:00:00.01 ALLOC TIME: 22:28:32 *
62 * PROGRAM NAME: IKFCBL00 REGION SIZE: 1024K ELAPSED TIME: 00:00:00.14 TCB TIME: 00:00:00.07 PROGRAM LOAD: 22:28:32 *
64 * CONDITION CODE: 00000 PERFORMANCE GROUP: 004
66 * JES2 CARDS: 11 SERVICE UNITS PAGES IN/OUT # SWAPS PAGES SWAP IN/OUT VIO PAGES IN/OUT *
68 * 477 0 / 0 0 0 / 0 0 / 0 *
70 *
72 * ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT *
74 * 180/D3380 0 191/D3390 6 191/D3390 6 191/D3390 12 191/D3390 6 191/D3390 9 *
76 *****
78 IEF236I ALLOC. FOR BUCHJOB GO COMP
80 IEF237I 191 ALLOCATED TO SYSLIN
82 IEF237I JES2 ALLOCATED TO SYSLOUT
84 IEF237I 180 ALLOCATED TO SYSLIB
86 IEF237I JES2 ALLOCATED TO SYSOUT
88 IEF142I BUCHJOB GO COMP - STEP WAS EXECUTED - COND CODE 0000
90 IEF285I SYS26144.T222832.RA000.BUCHJOB.LOADSET DELETED *------10
92 IEF285I VOL SER NOS= MYSTUF.
94 IEF285I JES2.JOB00752.S00103 SYSOUT
96 IEF285I SYS2.VSCLLIB KEPT *------63
98 IEF285I VOL SER NOS= PUB000.
100 IEF285I JES2.JOB00752.S00104 SYSOUT
102 IEF373I STEP /GO / START 26144.2228
104 IEF374I STEP /GO / STOP 26144.2228 CPU 0MIN 00.02SEC SRB 0MIN 00.00SEC VIRT 332K SYS 164K
106 *****
108 *
110 * STEP NUMBER: 2 USER CORE: 332K START TIME: 22:28:32 CPU TIME: 00:00:00.02 ACTIVE TIME: 00:00:00.05 *
112 * STEP NAME: GO SYSTEM CORE: 164K STOP TIME: 22:28:32 SRB TIME: 00:00:00.00 ALLOC TIME: 22:28:32 *
114 * PROGRAM NAME: LOADER REGION SIZE: 2048K ELAPSED TIME: 00:00:00.05 TCB TIME: 00:00:00.02 PROGRAM LOAD: 22:28:32 *
116 * CONDITION CODE: 00000 PERFORMANCE GROUP: 004
```

1	* JES2 CARDS: 0 SERVICE UNITS 477 PAGES IN/OUT 0 / 0 # SWAPS 0 PAGES SWAP IN/OUT 0 / 0 VIO PAGES IN/OUT 0 / 0 *										2
2	* * * * *										2
3	* ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT ADDR/UNIT I/O COUNT *										4
4	* 191/D3390 10 180/D3380 63 * * * * *										4
5	*****										6
6	IEF375I JOB /BUCHJOB / START 26144.2228										6
7	IEF376I JOB /BUCHJOB / STOP 26144.2228 CPU 0MIN 00.09SEC SRB 0MIN 00.01SEC										8
8											8
9											10
10											10
11											12
12											12
13											14
14											14
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```
2
4 1 22.28.32 MAY 24,1926
6
8 00001 IDENTIFICATION DIVISION.
00002 PROGRAM-ID. HASH.
00003 *
10 00004 ENVIRONMENT DIVISION.
00005 CONFIGURATION SECTION.
12 00006 SOURCE-COMPUTER. IBM-370.
00007 OBJECT-COMPUTER. IBM-370.
14 00008 *
00009 DATA DIVISION.
16 00010 WORKING-STORAGE SECTION.
00011 01 SS-TABLE.
18 00012 05 SS-NAME PIC X(25).
00013 05 SS-NO PIC S9(9).
20 00014 05 SS-DIV PIC S9(9) COMP-3 VALUE +1497.
00015 05 SS-TEMP PIC S9(9) COMP-3.
22 00016 05 SS-MAX-SIZE PIC S9(4) COMP SYNC VALUE +1500.
00017 05 SS-SUBSCRIPT PIC S9(9) COMP-3.
24 00018 05 SS-PERSON OCCURS 1500 INDEXED BY IP.
00019 10 SS-PERSON-NO PIC S9(9).
26 00020 10 SS-PERSON-NAME PIC X(25).
00021 *
28 00022 PROCEDURE DIVISION.
00023 A00-START.
30 00024 READY TRACE.
32 00025 * INITIALIZE THE TABLE TO ZERO
00026 PERFORM A10-ZERO-PERSON VARYING SS-SUBSCRIPT FROM 1 BY 1
00027 UNTIL SS-SUBSCRIPT = SS-MAX-SIZE.
34 00028 * ADD THREE PERSONS TO TABLE
00029 MOVE 123456 TO SS-NO.
36 00030 MOVE 'SIMON SULSER' TO SS-NAME.
00031 PERFORM C10-ADD-NAME.
38 00032
00033 MOVE 223344 TO SS-NO.
40 00034 MOVE 'HANS MUELLER' TO SS-NAME.
00035 PERFORM C10-ADD-NAME.
42 00036
00037 MOVE 676767 TO SS-NO.
44 00038 MOVE 'HEIDI MEIER' TO SS-NAME.
00039 PERFORM C10-ADD-NAME.
46 00040
00041 MOVE 654321 TO SS-NO.
48 00042 MOVE 'PETER MEIER' TO SS-NAME.
00043 PERFORM C10-ADD-NAME.
50 00044
00045 MOVE 852159 TO SS-NO.
52 00046 MOVE 'XENIA ALTDORF' TO SS-NAME.
00047 PERFORM C10-ADD-NAME.
54 00048
00049 * RETRIEVE A NAME WHICH IS NOT IN TABLE
56 00050 MOVE 111111 TO SS-NO.
00051 PERFORM C20-RETRIEVE-NAME.
58 00052 PERFORM A20-CHECK-FOUND.
00053
60 00054 * RETRIEVE A NAME WHICH IS IN TABLE
```

	2	HASH	22.28.32	MAY 24,1926	
2					2
4	00055	MOVE 223344 TO SS-NO.			4
	00056	PERFORM C20-RETRIEVE-NAME.			
6	00057	PERFORM A20-CHECK-FOUND.			6
	00058				
8	00059	* GO TO END OF PROGRAM			8
	00060	MOVE ZERO TO RETURN-CODE.			
10	00061	GO TO A99-STOP-RUN.			10
	00062				
12	00063	A10-ZERO-PERSON.			12
	00064	MOVE ZERO TO SS-PERSON-NO (SS-SUBSCRIPT).			
14	00065				14
	00066	A20-CHECK-FOUND.			
16	00067	IF SS-NAME = SPACES THEN			16
	00068	DISPLAY 'PERSON NICHT GEFUNDEN.'			
18	00069	ELSE			18
	00070	DISPLAY 'PERSON GEFUNDEN.'.			
20	00071				20
	00072	A99-STOP-RUN.			
22	00073	RESET TRACE.			22
	00074	STOP RUN.			
24					24
26					26
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4 00075 /
00076 C10-ADD-NAME.
6 00077 DIVIDE SS-NO BY SS-DIV GIVING SS-TEMP REMAINDER SS-SUBSCRIPT.
00078 ADD 1 TO SS-SUBSCRIPT.
8 00079 SET IP TO SS-SUBSCRIPT.
00080 SEARCH SS-PERSON
00081 AT END PERFORM C10-WRAP-AROUND
00082 WHEN SS-PERSON-NO (IP) = ZERO
12 00083 MOVE SS-NO TO SS-PERSON-NO (IP)
00084 MOVE SS-NAME TO SS-PERSON-NAME (IP)
14 00085 DISPLAY SS-NAME ' WRITTEN TO HASH TABLE.'.
00086
16 00087 C10-WRAP-AROUND.
00088 SET IP TO 1.
00089 SEARCH SS-PERSON
18 00090 WHEN IP = SS-SUBSCRIPT
20 00091 DISPLAY 'ERROR - SS-PERSON TABLE FULL. RUN TERMINATED.'
00092 DISPLAY 'INCREASE SS-PERSON- SS-DIV, SS-MAX-SIZE AND
22 00093 'RECOMPILE.'
00094 - MOVE 4 TO RETURN-CODE
24 00095 GO TO A99-STOP-RUN
00096 WHEN SS-PERSON-NO (IP) = ZERO
26 00097 MOVE SS-NO TO SS-PERSON-NO (IP)
00098 MOVE SS-NAME TO SS-PERSON-NAME (IP).
28 00099
00100 C20-RETRIEVE-NAME.
30 00101 DIVIDE SS-NO BY SS-DIV GIVING SS-TEMP REMAINDER SS-SUBSCRIPT.
00102 ADD 1 TO SS-SUBSCRIPT.
32 00103 SET IP TO SS-SUBSCRIPT.
00104 SEARCH SS-PERSON
34 00105 AT END PERFORM C20-WRAP-AROUND
00106 WHEN SS-PERSON-NO (IP) = ZERO
36 00107 MOVE SPACES TO SS-NAME
00108 WHEN SS-NO = SS-PERSON-NO (IP)
38 00109 MOVE SS-PERSON-NAME (IP) TO SS-NAME.
00110
40 00111 C20-WRAP-AROUND.
00112 SET IP TO 1.
42 00113 SEARCH SS-PERSON
00114 WHEN IP = SS-SUBSCRIPT
00115 MOVE SPACES TO SS-NAME
44 00116 WHEN SS-NO = SS-PERSON-NO (IP)
00117 MOVE SS-PERSON-NAME (IP) TO SS-NAME.
46 00118 * END PROGRAM HASH.
```

4	HASH	22.28.32	MAY 24,1926	
2				2
4				4
6	*STATISTICS*	SOURCE RECORDS = 118	DATA DIVISION STATEMENTS = 10	PROCEDURE DIVISION STATEMENTS = 59
8	*OPTIONS IN EFFECT*	SIZE = 2097152	BUF = 1048576	LINECNT = 57
10	*OPTIONS IN EFFECT*	SPACE1, FLAGW, SEQ, SOURCE		
12	*OPTIONS IN EFFECT*	NODMAP, NOPMAP, NOCLIST, NOSUPMAP, NOXREF, NOSXREF, LOAD, NODECK, APOST, NOTRUNC, NOFLOW		
14	*OPTIONS IN EFFECT*	NOTERM, NONUM, NOBATCH, NONAME, COMPILE=01, NOSTATE, NORESIDENT, NODYNAM, NOLIB, NOSYNTAX		
16	*OPTIONS IN EFFECT*	NOOPTIMIZE, NOSYMDMP, NOTEST, VERB, ZWB, SYST, NOENDJOB, NOLVL		
18	*OPTIONS IN EFFECT*	NOLST, NOFDECK, NOCDECK, LCOL2, L120, DUMP, ADV, NOPRINT,		
20	*OPTIONS IN EFFECT*	NOCOUNT, NOVBSUM, NOVBREF, LANGLVL(2)		
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VS LOADER

OPTIONS USED - PRINT, MAP, LET, CALL, RES, NOTERM, SIZE=307200, NAME=**GO

[illegible]

ENTRY	ADDRESS	FC28
16	TOTAL LENGTH	FC28
	ENTRY ADDRESS	9C010

2	A10-ZERO-PERSON (00001499),C10-ADD-NAME ,	2
	SIMON SULSER	
	WRITTEN TO HASH TABLE.	
4	C10-ADD-NAME ,	4
	HANS MUELLER	
	WRITTEN TO HASH TABLE.	
6	C10-ADD-NAME ,	6
	HEIDI MEIER	
	WRITTEN TO HASH TABLE.	
8	C10-ADD-NAME ,	8
	PETER MEIER	
	WRITTEN TO HASH TABLE.	
10	C10-ADD-NAME ,	10
	XENIA ALTDORF	
	WRITTEN TO HASH TABLE.	
12	C20-RETRIEVE-NAME ,A20-CHECK-FOUND ,	12
	PERSON NICHT GEFUNDEN.	
14	C20-RETRIEVE-NAME ,A20-CHECK-FOUND ,	14
	PERSON GEFUNDEN.	
16	A99-STOP-RUN ,	16
18		18
20		20
22		22
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BBBBBBBBBB	UU	UU	CCCCCCCC	HH	HH	JJJJJJJJ	0000000000	BBBBBBBBBB				
BBBBBBBBBB	UU	UU	CCCCCCCC	HH	HH	JJJJJJJJ	0000000000	BBBBBBBBBB				
BB	BB	UU	UU	CC	CC	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UU	UU	CC	HH	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UU	UU	CC	HH	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BB	BB	UU	UU	CC	HH	HH	HH	JJ	UU	UU	BB	BB
BBBBBBBBBB	UUUUUUUUUU	CCCCCCCC	HH	HH	HH	JJJJJJ	0000000000	BBBBBBBBBB				
BBBBBBBBBB	UUUUUUUUUU	CCCCCCCC	HH	HH	HH	JJJJJJ	0000000000	BBBBBBBBBB				

JJJJJJJJJ	77777777777	55555555555	222222222	AAAAAAAAA				
JJJJJJJJJ	77777777777	55555555555	22222222222	AAAAAAAAAAAA				
JJ	77	77	55	22	22	AA	AA	
JJ		77	55			22	AA	
JJ		77	55			22	AA	
JJ		77	55555555			22	AAAAAAAAAAAA	
JJ		77	555555555			22	AAAAAAAAAAAA	
JJ		77		55	22		AA	AA
JJ	JJ	77		55	22		AA	AA
JJ	JJ	77		55	22		AA	AA
JJJJJJJ	77	55555555555	22222222222				AA	AA
JJJJJJ	77	55555555555	22222222222				AA	AA

```
*****A      END      JOB      752      BUCHJOB      HASH TABLE      ROOM      10.28.32 PM 24 MAY 26      PRINTER1      SYS HMVS      JOB      752      END      A*****
```